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How to realize proof of concept with STM32 and its ecosystem

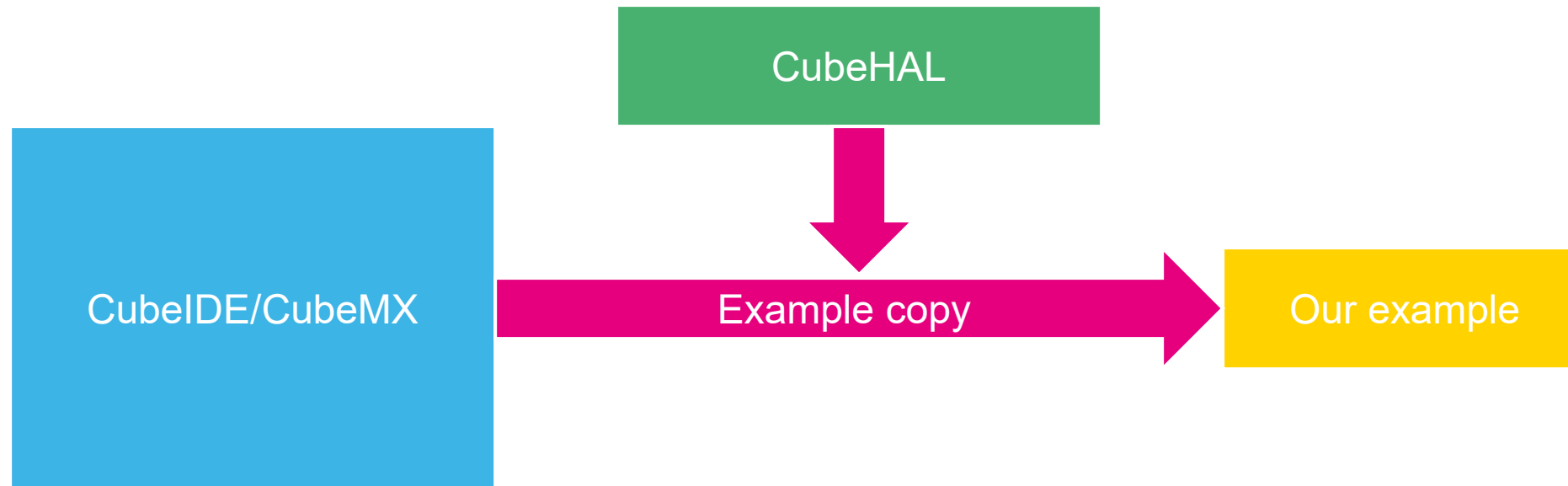


Purpose

- Allow to directly start from existing example
- Separate example automatically from CubeHAL package
 - Don't need copy use example from HAL
 - We not spend time with extracting example from CubeHAL

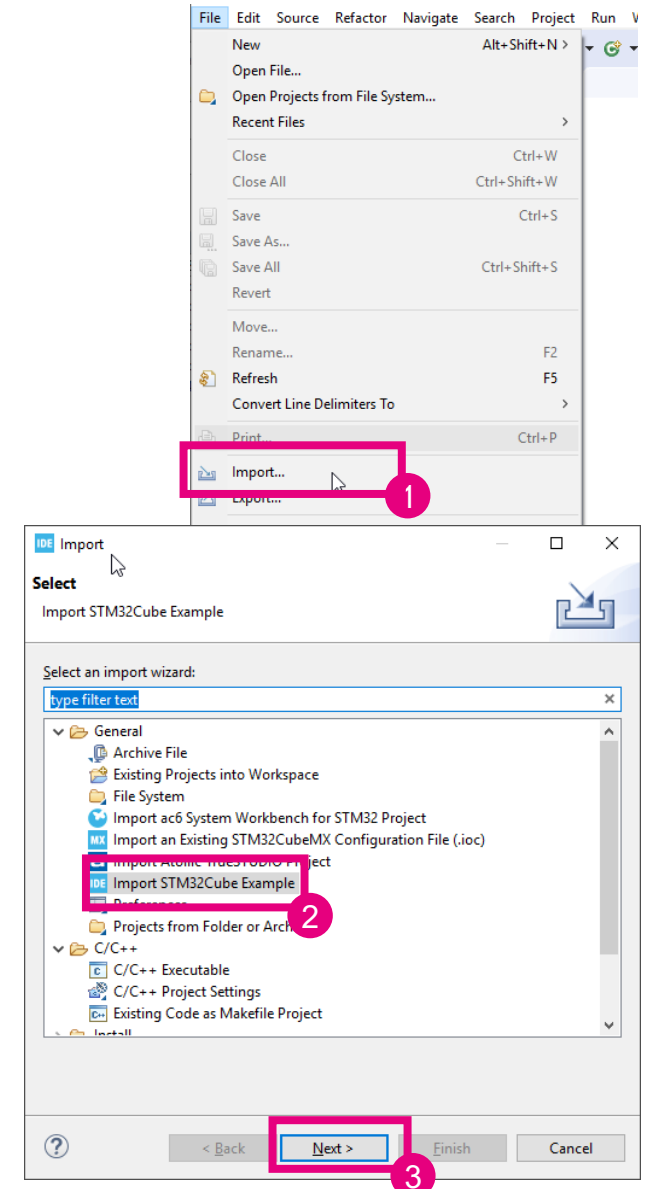
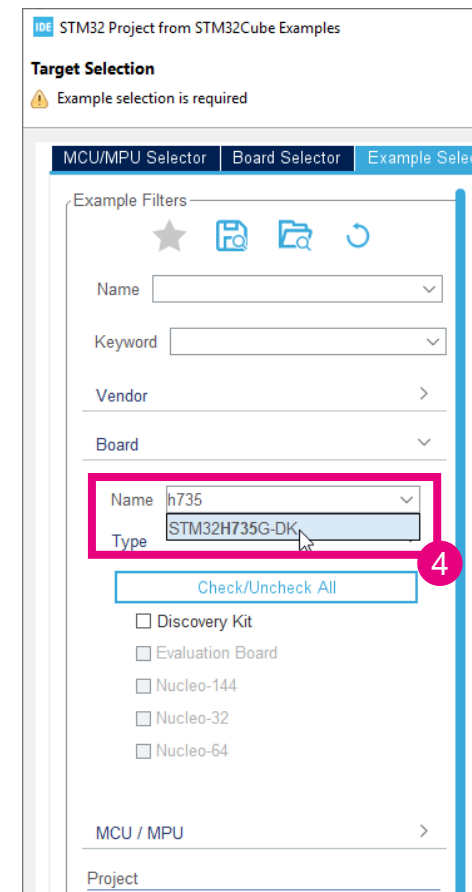
Start project as example

- STM32CubeMX or STM32CubeIDE allow to create project from example
- Project is no necessary to run from HAL package anymore
- Automatic copy on a new place



Fast start with Cube Examples

- File>Import 1
- Select Import STM32CubeExample 2
- Next 3
- Select STM32H735G-DK 4



Example types

- Example
 - Focused to use one IP
- Application
 - Use Library middleware
 - USB
 - RTOS
- Demonstration
 - Combine examples with applications

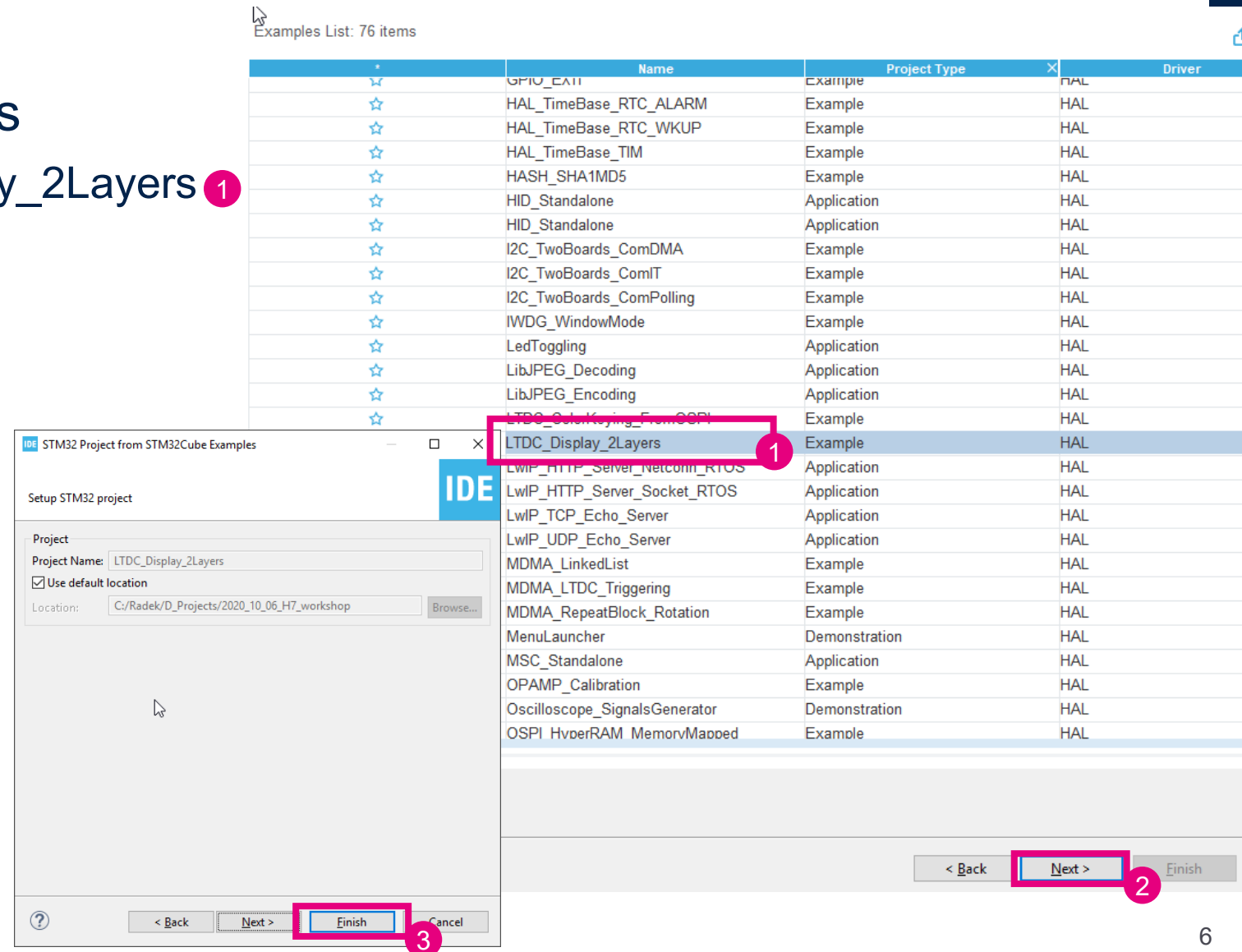
Examples List: 76 items

Export

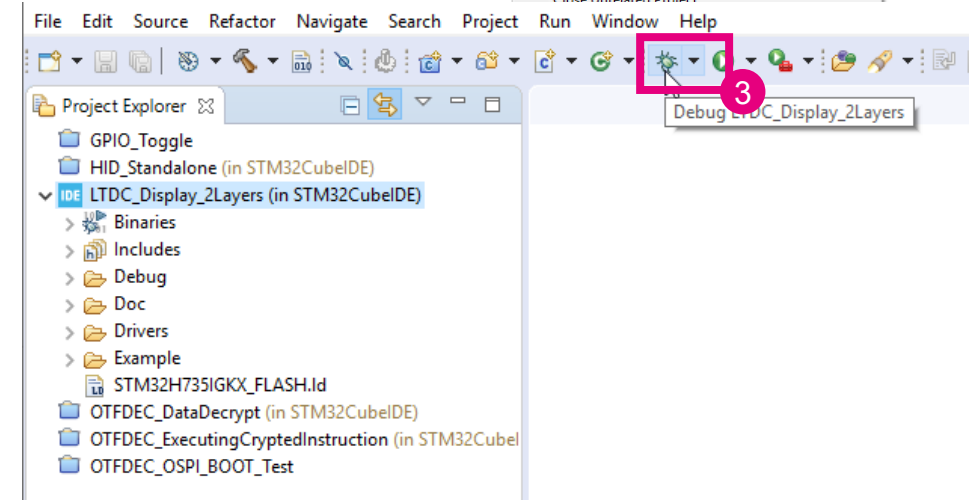
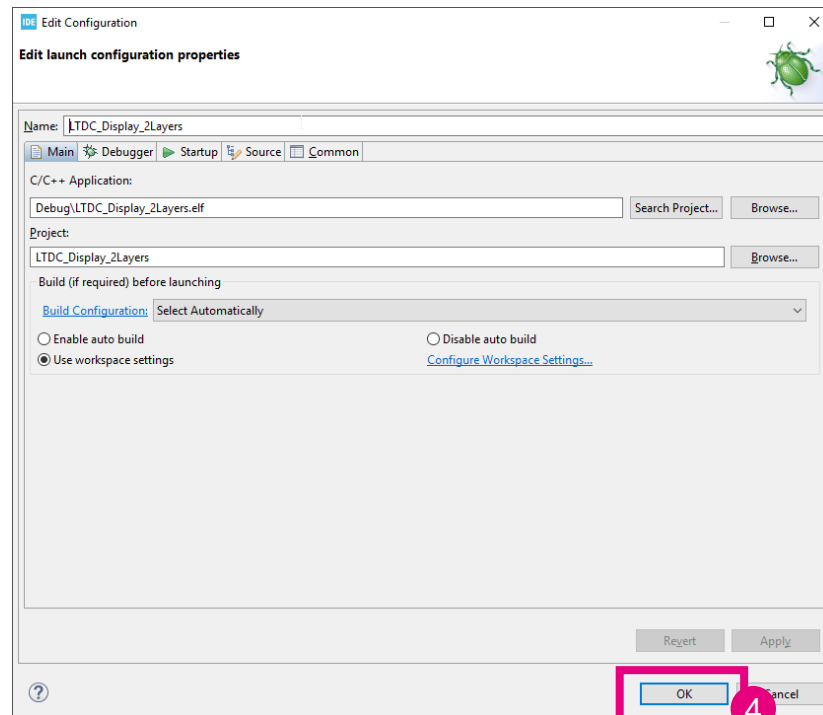
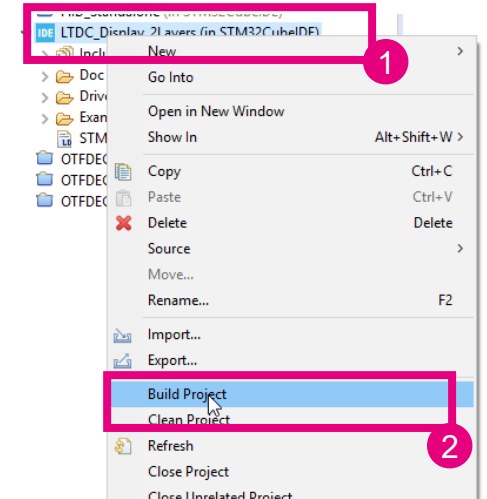
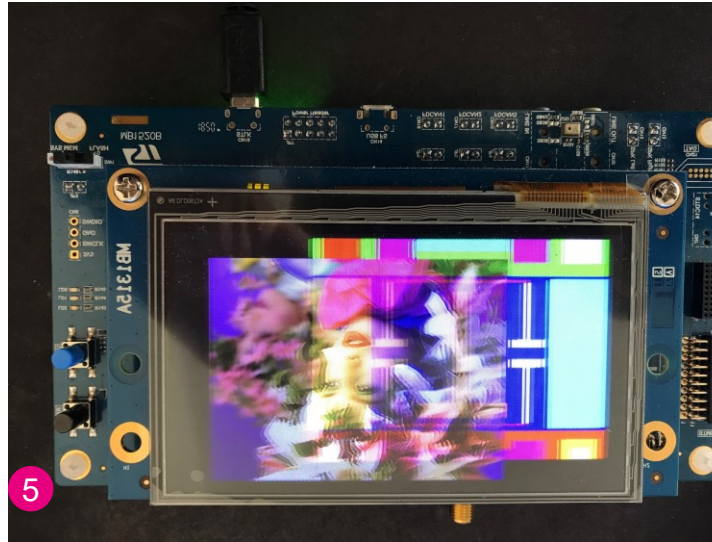
*	Name	Board	Board Type	Series	Project Type	Driver	CubeMX version
☆	ADC_AnalogWat...	STM32H735G-DK	Discovery Kit	STM32H7	Example	MAL	NA
☆	ADC_DMA_Trans...	STM32H735G-DK	Discovery Kit	STM32H7	Example	MAL	NA
☆	BSP	STM32H735G-DK	Discovery Kit	STM32H7	Example	MAL	NA
☆	ClockAndWeather	STM32H735G-DK	Discovery Kit	STM32H7	Application	MAL	NA
☆	COMP_Interrupt	STM32H735G-DK	Discovery Kit	STM32H7	Example	MAL	NA
☆	CORDIC_Sin_DMA	STM32H735G-DK	Discovery Kit	STM32H7	Example	MAL	NA
☆	CRC_Data_Rever...	STM32H735G-DK	Discovery Kit	STM32H7	Example	MAL	NA
☆	CRC_Example	STM32H735G-DK	Discovery Kit	STM32H7	Example	MAL	NA
☆	CRC_UserDefine...	STM32H735G-DK	Discovery Kit	STM32H7	Example	MAL	NA
☆	CRYPT_AES_GCM	STM32H735G-DK	Discovery Kit	STM32H7	Example	MAL	NA
☆	Crypto_Selftest	STM32H735G-DK	Discovery Kit	STM32H7	Application	MAL	NA
☆	DAC_SignalsGen...	STM32H735G-DK	Discovery Kit	STM32H7	Example	MAL	NA
☆	DFSDM_AudioRe...	STM32H735G-DK	Discovery Kit	STM32H7	Example	MAL	NA
☆	DFU_Standalone	STM32H735G-DK	Discovery Kit	STM32H7	Application	MAL	NA
☆	DMA2D_Blending...	STM32H735G-DK	Discovery Kit	STM32H7	Example	MAL	NA
☆	DMA2D_Memory...	STM32H735G-DK	Discovery Kit	STM32H7	Example	IL	NA
☆	DMA2D_MemTo...	STM32H735G-DK	Discovery Kit	STM32H7	Example	MAL	NA
☆	DMA2D_MemTo...	STM32H735G-DK	Discovery Kit	STM32H7	Example	MIX	NA
☆	DMA2D_MemTo...	STM32H735G-DK	Discovery Kit	STM32H7	Example	MAL	NA
☆	DMAMUX_Reque...	STM32H735G-DK	Discovery Kit	STM32H7	Example	MAL	NA
☆	DTS_GetTemper...	STM32H735G-DK	Discovery Kit	STM32H7	Example	MAL	NA
☆	ExtMem_Boot	STM32H735G-DK	Discovery Kit	STM32H7	Application	MAL	NA
☆	FatFs_uSD_RTOS	STM32H735G-DK	Discovery Kit	STM32H7	Application	MAL	NA
☆	FatFs_uSD_Stan...	STM32H735G-DK	Discovery Kit	STM32H7	Application	MAL	NA
☆	FDCAN_Loopback	STM32H735G-DK	Discovery Kit	STM32H7	Example	MAL	NA
☆	FLASH_ErasePr...	STM32H735G-DK	Discovery Kit	STM32H7	Example	MAL	NA
☆	FMAC_FIR_DMA...	STM32H735G-DK	Discovery Kit	STM32H7	Example	MAL	NA

Select the Example

- Possible to see of HAL examples
 - For test we can select LTDC_Display_2Layers ①
- Next ②
- Finish ③

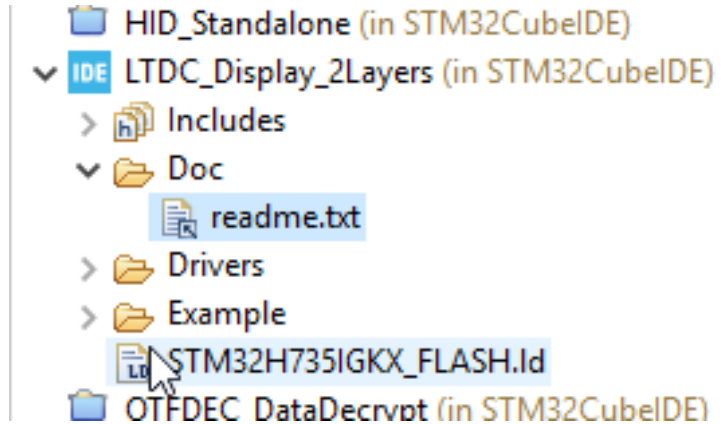


- Build project
 - Right click on project ①
 - Build ②
- Run debug ③
- Confirm debug configuration ④
- Run example F8 ⑤



About examples

- Information are in Doc folder
- Readme file
 - Example description
 - What is used in example
 - How to configure board for example (Jumpers, ..)
 - Additional SW needed
 - How to run it



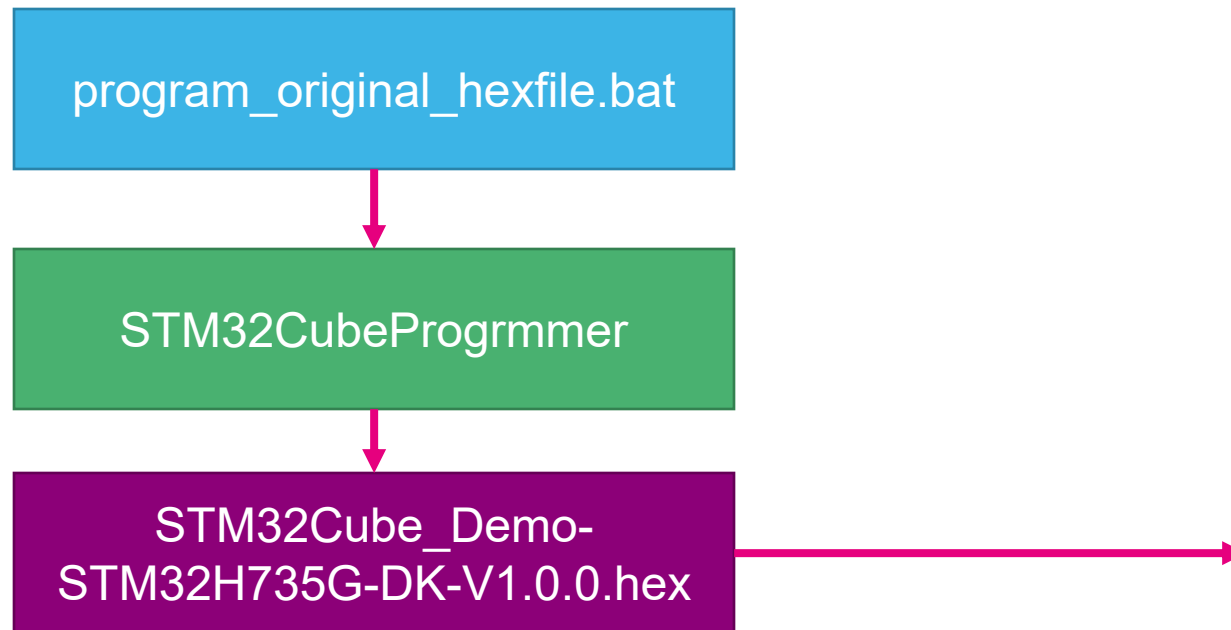
```
1 @par Example Description
2
3 How to configure the LTDC peripheral to display two layers at the same time.
4 The goal of this example is to explain how to blend two layers and how to display
5 an image with L8 (8bits per pixels indexed) as pixel format.
6
7 At the beginning of the main program the HAL_Init() function is called to reset
8 all the peripherals, initialize the Flash interface and the systick.
9 The SystemClock_Config() function is used to configure the system clock for STM32H735XX Devices :
10 The CPU at 520 MHz.
11 The HCLK for D1 Domain AXI and AHB3 peripherals , D2 Domain AHB1/AHB2 peripherals and D3 Domain AHB4 peripherals at 520 MHz/2.
12 The APB clock dividers for D1 Domain APB3 peripherals, D2 Domain APB1 and APB2 peripherals and D3 Domain APB4 peripherals to run at 520 MHz/4.
13
14 After LCD initialization, the LCD layer 1 and Layer 2 are configured as following :
15 - Layer 1 is configured to display an image loaded from flash memory with direct
16   color (RGB565) as pixel format and 320x240 (QVGA) size.
17 - Layer 2 is configured to display an image loaded from flash memory with indirect
18   color (L8) as pixel format and 320x240 (QVGA) size.
19 To display an image with an indirect color as pixel format a color lookup table (CLUT)
20 is loaded, then every byte from image data (L8_320x240) is considered as
21 a position in color lookup table (CLUT).(see example below)
22
23
```

Conclusion

- Quick and simple start to check proof of concept with STM32Cube examples
- Start directly from example
- Don't need manually copy example from HAL
- We not spend time with extracting example from CubeHAL

Restore original demo

- In folder TOOLS execute **program_original_hexfile.bat**
- It will erase the board and upload original demo
- The demo is very big and erase/programming take long time



Thank you

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